

Problem Set 1

1. Sleep.c

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sleep 10
$
```

Shell correctly waited 10 seconds before moving to the next instruction (\$). I don't know why test failed.

2. Sixfive.c

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo "5 100 18 6 xv6 hello 30-25" > sixfive.txt
$ sixfive sixfive.txt
5
100
18
6
6
30
25
$
```

3. Memdump.c

```
$ cat > input.txt
deadc0de
$ memdump hhcccc < input.txt
25956
25697
c
0
d
e
$
```

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4. Find.c

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ |
```

5. Find modified Exec.c

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ find . wc
./wc
$ find . wc -exec echo hello
hello ./wc
```

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6. Make grade:

Exec modification of find fails the test because I did not modify find.c instead created a new program exec.c so that I don't mess up the working find.c and keep separate clear concepts.

```

marium@DESKTOP-0K07MJA: x + v
make[1]: Leaving directory '/home/marium/xv6-labs-2025'
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (5.9s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (0.9s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.0s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (1.2s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (1.0s)
(Old xv6.out.sixfive_readme failure log removed)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: FAIL (0.9s)
...
init: starting sh
$ sixfive sixfive.txt README
GOOD 5
GOOD 100
GOOD 18
GOOD 6
$ qemu-system-riscv64: terminating on signal 15 from pid 1529 (make)
MISSING: ^1810^
MISSING: ^1810^
QEMU output saved to xv6.out.sixfive_readme
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.0s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.3s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.0s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (0.9s)
== Test find, recursive ==
$ make qemu-gdb
QEMU output saved to xv6.out.sixfive_readme
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.0s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.3s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.0s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (0.9s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.2s)
== Test exec ==
$ make qemu-gdb
exec: OK (0.9s)
(Old xv6.out.find_exec failure log removed)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (0.9s)
(Old xv6.out.find_exec_args failure log removed)
== Test exec, recursive find ==
$ make qemu-gdb
exec, recursive find: OK (1.4s)
(Old xv6.out.test_find_sh failure log removed)
== Test time ==
time: OK
Score: 121/131
make: *** [Makefile:360: grade] Error 1
marium@DESKTOP-0K07MJA:~/xv6-labs-2025$ nano user/sixfive.c
marium@DESKTOP-0K07MJA:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
riscv64-unknown-elf-gcc -Wall -Werror -D -fno-onit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-built-in-stdlib -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcap -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sixfive.o user/sixfive.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_sixfive user/sixfive.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_sixfive > user/sixfive.asm
riscv64-unknown-elf-objdump -t user/_sixfive | sed '1,/SYMBOL TABLE/d; s/ / /; /%d/' > user/sixfive.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/cat user/echo user/forktest user/grep user/init user/kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_exec user/_memdump

```